

CBSE Class 10 Mathematics — Statistics

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains *questions. All questions are compulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The mean of the first 5 natural numbers is:
(A) (A) 2
(B) (B) 3
(C) (C) 4
(D) (D) 5
2. The mode of the data set 2, 3, 4, 3, 5, 3, 6, 3 is:
(A) (A) 2
(B) (B) 4
(C) (C) 3
(D) (D) 5
3. If the mean of 5 observations $x, x + 2, x + 4, x + 6, x + 8$ is 11, then the value of x is:
(A) (A) 7
(B) (B) 8
(C) (C) 9
(D) (D) 6
4. The median of the data 7, 3, 5, 9, 1 is:
(A) (A) 3
(B) (B) 5
(C) (C) 7
(D) (D) 1
5. For a distribution, if mean = 20 and median = 21, then the mode is:
(A) (A) 22

- (B) (B) 24
- (C) (C) 21
- (D) (D) 23

6. Assertion (A): The mean of the first five natural numbers is 3. Reason (R): The mean of a set of observations is calculated by dividing the sum of observations by the total number of observations.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): The mode of the data 2, 3, 4, 3, 2, 3 is 3. Reason (R): Mode is the value that appears most frequently in a data set.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): Median of the data 5, 1, 3, 2, 4 is 3. Reason (R): Median is the middle value when data is arranged in ascending or descending order.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): The range of the data 10, 15, 20, 25, 30 is 20. Reason (R): Range is defined as the difference between the maximum and minimum values of the data.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): The sum of deviations of the observations x_i from their mean $\bar{x}$ is 0.
Reason (R): $\sum(x_i - \bar{x}) = 0$ is a fundamental property of the arithmetic mean.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

- Find the mean of the first n natural numbers.
- If the mean of the following distribution is 6, find the value of p : $x : 2, 4, 6, 10, p + 5$; $f : 3, 2, 3, 1, 2$.
- The median of a distribution is 15 and the mean is 12. Find the mode using the empirical relationship.
- Find the class mark of the class interval $10 - 25$ and $35 - 50$.
- If $\sum f_i = 20$ and $\sum f_i x_i = 2p + 120$, find p if the mean is 7.

Section D: Long Answer (4-5 marks each)

- The mean of the following frequency distribution is 50. Find the value of p if the sum of frequencies is 100. — Class — 0-20 — 20-40 — 40-60 — 60-80 — 80-100 —
————— Frequency — 17 — p — 32 — p — 19 —
- The following table gives the daily income of 50 workers of a factory. Find the mean and daily income of the workers.*
120|120 – 140|140 – 160|160 – 180|180 – 200|| — — — | — — — | — — — | — — — | — — —
| — — — ||Number of workers|12|14|8|6|10|
- Find the median of the following data: — Class interval — 0-10 — 10-20 — 20-30 —
30-40 — 40-50 — ————— — Frequency — 5 — 15 — 20 — 7
— 3 —
- Find the modal class and the mode for the following distribution: — Marks — 0-10
— 10-20 — 20-30 — 30-40 — 40-50 — ————— — Number of
students — 4 — 10 — 16 — 12 — 8 —
- If the median of a distribution is 28 and the mean is 27, find the mode using the empirical relationship.
- The following distribution shows the daily pocket allowance of 50 children of a locality. If the mean pocket allowance is {Rs. }18, find the missing frequency f for the class interval $19 - 21$: *Daily allowance (in Rs.) : 11 – 13, 13 – 15, 15 – 17, 17 – 19, 19 – 21, 21 – 23, 23 – 25* *Number of children : 7, 6, 9, 13, f, 5, 4*

7. The annual profits earned by 30 shops of a shopping complex in a locality are given below: Profit (in lakhs): 0-10, 10-20, 20-30, 30-40, 40-50, 50-60 Number of shops: 5, 8, 6, 4, 3, 4 Draw both 'less than' and 'more than' ogives for the given data on the same graph and find the median from the intersection point of the ogives.
8. If the median of the following frequency distribution is 28.5, find the values of x and y if the total frequency is 60: Class Interval: 0-10, 10-20, 20-30, 30-40, 40-50, 50-60 Frequency: 5, x , 20, 15, y , 5
9. The frequency distribution of the heights of 50 students is given below. Find the modal height of the students: Height (cm): 140-145, 145-150, 150-155, 155-160, 160-165 Number of students: 8, 12, 15, 10, 5